

```
CREATE DATABASE smart_home_energy;

USE smart_home_energy;

CREATE TABLE rooms (
    room_id INT AUTO_INCREMENT PRIMARY KEY,
    room_name VARCHAR(50)
);

CREATE TABLE devices (
    device_id INT AUTO_INCREMENT PRIMARY KEY,
    device_name VARCHAR(100),
    room_id INT,
    status VARCHAR(20),
    FOREIGN KEY (room_id) REFERENCES rooms(room_id)
);

CREATE TABLE energy_logs (
    log_id INT AUTO_INCREMENT PRIMARY KEY,
    device_id INT,
    energy_kwh DECIMAL(10,2),
    log_date DATE,
    FOREIGN KEY (device_id) REFERENCES devices(device_id)
);

INSERT INTO rooms (room_name)
VALUES ('Living Room'), ('Kitchen'), ('Bedroom');

INSERT INTO devices (device_name, room_id, status)
VALUES
('Fan', 1, 'ON'),
('Refrigerator', 2, 'ON'),
('Light', 3, 'OFF');

INSERT INTO energy_logs (device_id, energy_kwh, log_date)
```

```
VALUES
(1, 2.5, '2026-05-13'),
(2, 5.2, '2026-05-13'),
(3, 1.1, '2026-05-13');

INSERT INTO energy_logs (device_id, energy_kwh, log_date)
VALUES (1, 3.0, '2026-05-14');

SELECT * FROM energy_logs;

UPDATE devices
SET status = 'OFF'
WHERE device_id = 1;

DELETE FROM energy_logs
WHERE log_id = 1;

DELIMITER //
```

```
CREATE PROCEDURE DailyEnergyUsage()
BEGIN
    SELECT r.room_name,
           e.log_date,
           SUM(e.energy_kwh) AS total_energy
    FROM energy_logs e
    JOIN devices d ON e.device_id = d.device_id
    JOIN rooms r ON d.room_id = r.room_id
    GROUP BY r.room_name, e.log_date;
END //
```

```
DELIMITER ;

CALL DailyEnergyUsage();
```

MySQL Workbench

MySQL Model x localhost x

File Edit View Query Database Server Tools Scripting Help

Query 1 x SQL File 2* SQL File 5*

Limit to 1000 rows

```
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2 USE smart_home_energy;
3 CREATE TABLE rooms (
4     room_id INT AUTO_INCREMENT PRIMARY KEY,
5     room_name VARCHAR(50)
6 );
7 CREATE TABLE devices (
8     device_id INT AUTO_INCREMENT PRIMARY KEY,
9     device_name VARCHAR(100),
10    room_id INT,
11    status VARCHAR(20),
12    FOREIGN KEY (room_id) REFERENCES rooms(room_id)
13 );
```

Result Grid Filter Rows: Export: Wrap Cell Contents: I

room_name	log_date	total_energy
Kitchen	2026-05-13	5.20
Bedroom	2026-05-13	1.10
Living Room	2026-05-14	3.00

Result 2 x Read Only

15:28 13-05-2026

MySQL Workbench

MySQL Model x localhost x

File Edit View Query Database Server Tools Scripting Help

Query 1 x SQL File 2* SQL File 5*

Limit to 1000 rows

```
10 room_id INT,
11 status VARCHAR(20),
12 FOREIGN KEY (room_id) REFERENCES rooms(room_id)
13 );
14 CREATE TABLE energy_logs (
15     log_id INT AUTO_INCREMENT PRIMARY KEY,
16     device_id INT,
17     energy_kwh DECIMAL(10,2),
18     log_date DATE,
19     FOREIGN KEY (device_id) REFERENCES devices(device_id)
20 );
21 INSERT INTO rooms (room_name)
22 VALUES ('Living Room'), ('Kitchen'), ('Bedroom');
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Result Grid Filter Rows: Export: Wrap Cell Contents: I

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MySQL Workbench

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Result Grid

room_name	log_date	total_energy
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Result 2 x

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15:28 13-05-2026